

A Project Report On

“Development of website for sigh language translator”

SUBMITTED TO

Pimpri Chinchwad Education Trust’s

Pimpri Chinchwad College Of Engineering ,[PCCOE]

(An Autonomous Institute)

NBA Accredited, NAAC Accredited with 'A' Grade, AICTE Approved

ISO 9001:2015 Certified | Permanently Affiliated to SPPU, Pune

FOR FULFILLMENT OF

THE ENGINEERING CHEMISTRY LAB OF

F.Y.B.TECH.

SUBMITTED BY

Sumit Jadhav (PRN:125B1B074)

Nilesh Zende (PRN:125B1B096)

Tanuja Kamble (PRN:125B1B082)

Om Atole (PRN:125B1B076)

UNDER THE GUIDANCE

OF

DR. SANDIP JAGADALE

DEPARTMENT OF APPLIED SCIENCE AND HUMANITIES

PIMPRI CHINCHWAD COLLEGE OF ENGINEERING, NIGDI, PUNE-411062

AY 2025-2026 (Sem I)

CERTIFICATE

This is to certify that the project entitled

“Development of website for sigh language translator”

is the bonafide work of Sumit Jadhav (PRN:125B1B074), Nilesh Zende (PRN:125B1B096), Tanuja Kamble (PRN:125B1B082), Om Atole (PRN:125B1B076) submitted to Pimpri Chinchwad College Of Engineering, Pune In the fulfilment of the requirement for the Engineering Chemistry Project activity of “First Year Bachelor of Technology (Computer Engineering) ".

Date:

Project Guide

Dr. Sandip Jagadale

DECLARATION OF CANDIDATES

We, the undersigned solemnly declare that the project report

"**DEVLOPMENT OF WEBSITE FOR SIGN LANGUAGE TRANSLATOR"** Is based on a technological application carried out by us during the course of our study under the guidance of Dr. Sandip Jagadale. We assert the statement made and conclusion drawn is the outcome of our tech ideology work whenever we used materials (programming and text) from other sources, it is indicated in the reference and acknowledge.

Sumit Jadhav (PRN:125B1B074)
Nilesh Zende (PRN:125B1B096)
Tanuja Kamble (PRN:125B1B082)
Om Atole (PRN:125B1B076)

Certified By

……..

Dr. Sandip Jagadale

Project Guide

Acknowledgment

In the present world of competition there is a rare existence in which those who have the will to come forward succeed. First of all, we would like to thank the supreme power the Almighty God who is obviously the one who has always guided us to work on the right path of life. Without his grace this project could not become a reality.

We would like to express our sincere thanks to the Management of PCCOE, Director, the HOD, and Dept. of AS&H for their constant motivation. We are thankful to all our teachers and friends who have been always helping and encouraging me throughout the semester.

We would like to sincerely express our deep sense of gratitude to our guide Dr. Sandip Jagadale Sir, Dept. of Applied Science and Humanity, Pimpri Chinchwad College of Engineering, Pune for his extraordinary cooperation, invaluable guidance and supervision. This report is the result of his generous attitude.

We express our deepest sense of gratitude to our parents, for their deep- rooted love and affection, blessing prayers, sacrifice, tender care, spiritual support that paved the way towards achieving the goals.

Sumit Jadhav (PRN:125B1B074)
Nilesh Zende (PRN:125B1B096)
Tanuja Kamble (PRN:125B1B082)
Om Atole (PRN:125B1B076)

ABSTRACT

Sign language is a **complex visual-manual** form of communication that serves as a primary language for many deaf and hard-of-hearing individuals, yet its reach and utility have grown far beyond this community in the modern world.

Recent advancements in real-time technology have enabled automated systems to recognize and translate sign language into text or speech, greatly enhancing communication accessibility and autonomy for sign language users. These innovations utilize **deep learning, computer vision, and artificial intelligence** to accurately interpret signs from live video, making real-time, reliable communication possible in diverse environments including public services, education, and personal interactions .

In addition to serving as a bridge for the hearing-impaired, sign language facilitates early communication for children, enriches social inclusion, and provides an alternative means of interaction in situations where spoken language is impractical, such as noisy or silent environments. Modern real-time applications of sign language, including **gesture-to-text and gesture-to-speech technologies**, not only break down barriers but also support inclusive education, emergency response, and remote communication, demonstrating the **versatile impact of sign language in today's technologically advanced society.**

These tools empower people with hearing or speech challenges by giving them greater independence and participation in everyday activities, contributing to a more inclusive and connected modern world.

INDEX

SR.NO	CONTENT
1	**INTRODUCTION**
2	**WEBSITE OVERVIEW(USE,OBJECTIVE,SCOPE)**
3	**APPLICATIONS IN MODERN WORLD**
4	**ADVANTAGES**
5	**USE OF GESTURES AND AUDIO**
6	**CONCLUSION**

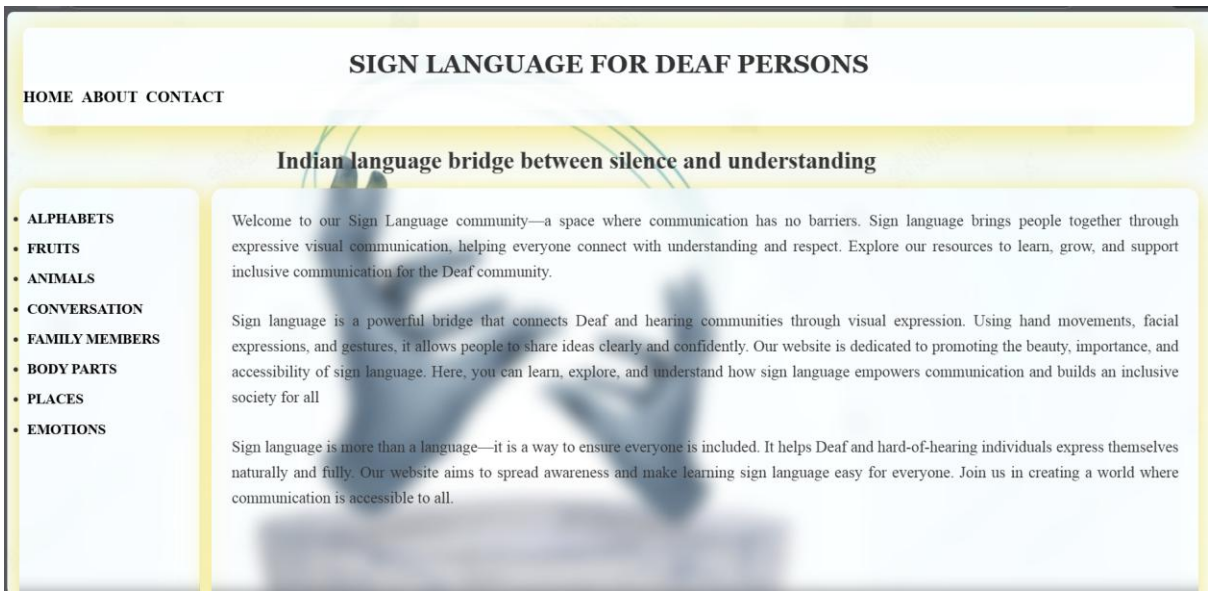
INTRODUCTION

This website is made to help people learn sign language in a simple and friendly way. It shows each sign using a clear image and also plays the voice audio of the related word.

The user can click on a picture and see how the sign is done step by step. At the same time, the user can hear how the word is spoken, so it is easy to connect the gesture with the sound. This makes learning comfortable for beginners and students.

The website is helpful for people who want to communicate better with deaf or hard-of-hearing persons. It can also support normal classroom learning or self-study at home. Because it is on the web, learners can access it anytime and from anywhere, using a phone, tablet, or computer.

By putting sign language on a website, it helps remove communication barriers for deaf and hard-of-hearing users and allows hearing people to build better communication skills and awareness. Online access also means users can learn anytime and anywhere, which makes learning flexible, comfortable, and suitable for different learning speeds and styles.



WEBSITE OVERVIEW

Use of the website:

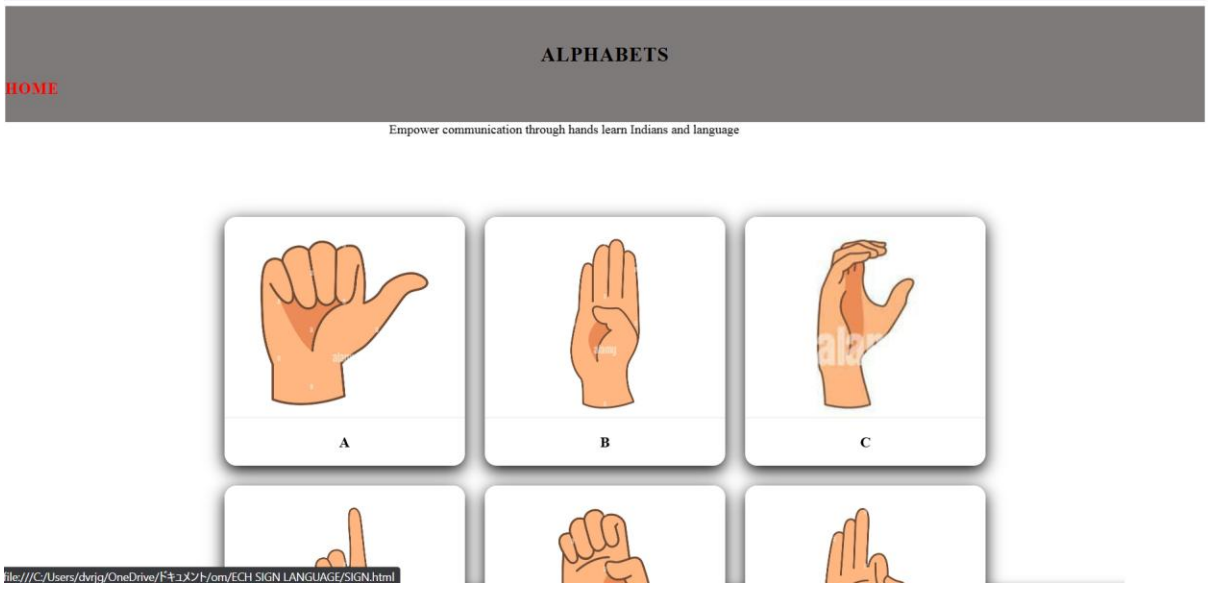
It helps users see each sign clearly through images, so they can copy the correct hand shape and movement. It lets users hear the word or sentence through audio, so they can connect spoken language with the sign. It supports self-learning at home, in class, or on mobile, without needing a teacher all the time.

Objectives of the website:

To teach basic sign language in a simple and visual way to beginners. To make communication easier between hearing people and deaf or hard-of-hearing people. To provide an easy tool for students, parents, and teachers to quickly learn and revise signs.

Scope of the website:

It can be used by school and college students, teachers, parents, and anyone interested in sign language. More signs, categories (like alphabets, numbers, daily words, sentences), and languages can be added in the future. The website can later include quizzes, videos, and interactive practice so users can test their knowledge and track progress.



APPLICATION IN MODERN WORLD

Communication is the foundation of understanding in between people there thoughts, ideas and information.

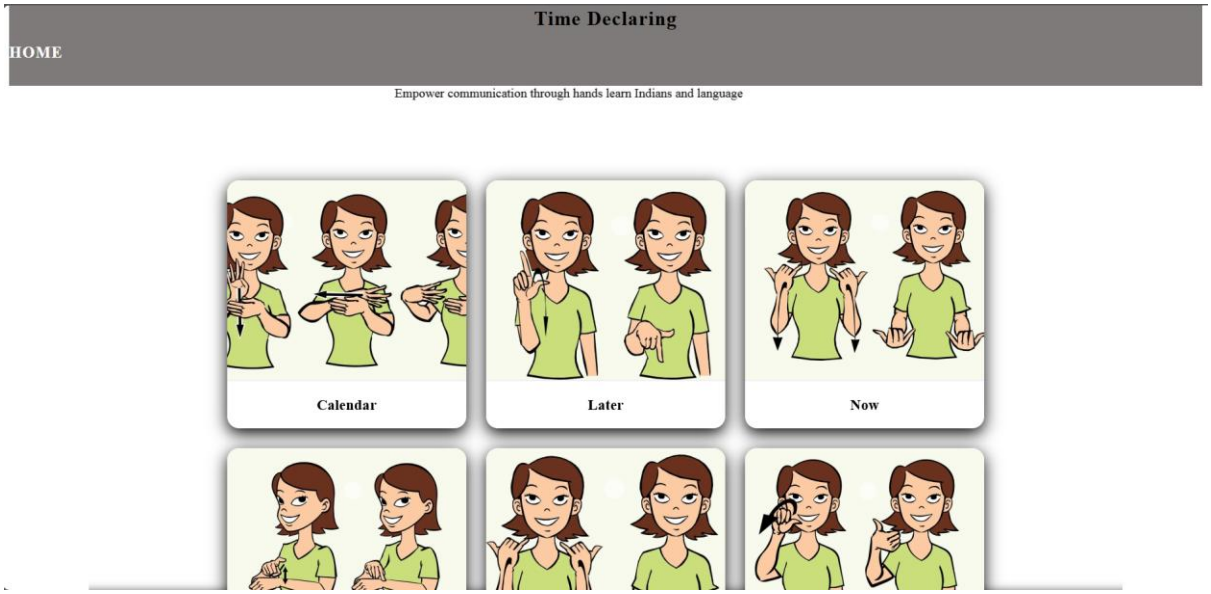
This helps to share there feelings from one person to another, avoid confusion and misunderstanding.

To express this important key feature of human beings we have designed this website ,because people who are deaf they also need the similar type of platform were they can also understand and express there feelings.

Application of our website in modern world is that when we face difficulty while talking with the deaf people this helps to sort out the things a little bit and make the easier way to communicate to them in a more effective and stronger way.

This website has included with much of the words that are helpful in communicating to the deaf people .words are presented in the picture format as well as are accessible in voice mode

By this, people who went to talk to deaf people or who are not able to empress there communication to people are able to convey there message.



ADVANTAGES

Advantages of our deaf person sign language website are like it enhanced visual communication with the help of sign language image, captions and voice given to the images. Website helps to people learn sign language and with the help of this sign language website they communicate properly with deaf person and to make strong communication. Learning is simple in our website with beautiful images and funny character .

This website only prepare for learning without districtions and without any unwanted add. Our website provides different sections with images for different communication types like Alphabet declaring, basic conversion declaring, Animal declaring, fruits declaring, time and family members declaring images.

Our website has simple structure and the use of website is easy and not need any experience of handling website so this is most useful website to learn the deaf person sign language. So many websites are available but they not provide proper learning structure but our website provides proper learning structure. This website is free for all the users and available in free of cost and it's name is Indian Sign Language

Overall, this website serves as a powerful tool to democratize sign language education, promote accessibility, and empower users to communicate effectively across diverse modern settings.

USE OF GESTURE AND AUDIO

Use of gestures :

Images of hand shapes and body movements help learners clearly see how to make each sign, which is important because sign language is a visual language.

Connecting a picture of a gesture with a word or meaning improves memory, because learners use both their eyes and their body movement when they copy the sign.

Visual signs support people who struggle with spoken language, young children, or deaf and hard-of-hearing users, so the website becomes more inclusive.

Use of audio :

Audio lets learners hear the correct pronunciation of the word or sentence that matches each sign, which helps hearing users improve their speech and listening skills.

Listening while watching the sign connects spoken language with sign language, which can strengthen vocabulary and understanding in both languages.

Audio support is useful for parents, teachers, and hearing friends who want to communicate better with sign language users, because they can listen and learn at the same time.

The use of gestures combined with audio in your sign language website provides a powerful and comprehensive learning experience with several important benefits

CONCLUSION

This sign language website stands as a simple yet powerful tool that brings learning to life for everyone. By combining clear images of hand gestures with matching voice audio, it turns complex signs into easy steps that anyone can follow whether you're a student practicing at home, a parent helping your child communicate, a teacher in a classroom, or just someone curious about connecting with deaf friends.

No more guessing or struggling alone; users click, see the exact movement, hear the word, and start signing right away. This hands-on approach makes lessons stick longer and feel more natural, turning beginners into confident communicators in just a few sessions.

What makes this website truly special is how it builds bridges in a divided world. Sign language isn't just for the deaf community it's a skill that opens doors for hearing people too, reducing misunderstandings and fostering real inclusion. Imagine a classroom where every child understands each other, or a family where everyone shares stories without barriers.